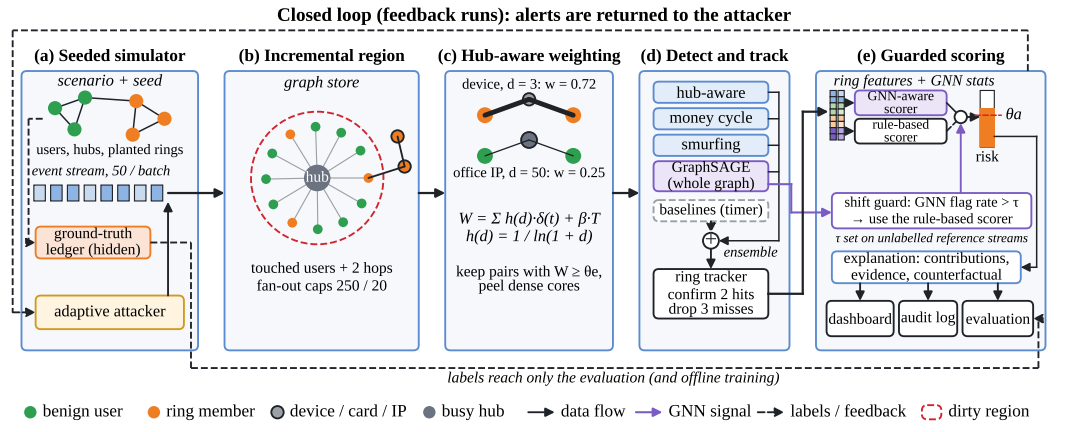

Closed loop (feedback runs): alerts are returned to the attacker
(a) Seeded simulator
scenario + seed
users, hubs, planted rings
event stream, 50 / batch
ground-truth ledger (hidden)
adaptive attacker
(b) Incremental region
graph store
hub
touched users + 2 hops
fan-out caps 250 / 20
(c) Hub-aware weighting
device, d = 3: w = 0.72
office IP, d = 50: w = 0.25
W = Σ h(d)·δ(t) + β·T
h(d) = 1 / ln(1 + d)
keep pairs with W ≥ θe,
peel dense cores
(d) Detect and track
hub-aware
money cycle
smurfing
GraphSAGE (whole graph)
baselines (timer)
ensemble
ring tracker
confirm 2 hits
drop 3 misses
(e) Guarded scoring
ring features + GNN stats
GNN-aware scorer
rule-based scorer
θa
risk
shift guard: GNN flag rate > τ
→ use the rule-based scorer
τ set on unlabelled reference streams
explanation: contributions, evidence, counterfactual
dashboard
audit log
evaluation
labels reach only the evaluation (and offline training)
benign user
ring member
device / card / IP
busy hub
data flow
GNN signal
labels / feedback
dirty region